

Step 1: Define the Matrices

From your input:

Matrix A contains:

$$A = \begin{bmatrix} 5 & 10 & 15 & 20 \\ 5 & 10 & 15 & 20 \\ 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix}$$

Matrix B contains:

$$B = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}$$

Step 2: Partition the Matrices for Strassen's Method

Strassen's method works best for matrices of size $2^n \times 2^n$. Here we have 4×4 matrices, so we partition them into 2×2 submatrices.

Partition Matrix A:

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$$

Where:

$$A_{11} = \begin{bmatrix} 5 & 10 \\ 5 & 10 \end{bmatrix}, \quad A_{12} = \begin{bmatrix} 15 & 20 \\ 15 & 20 \end{bmatrix}$$
$$A_{21} = \begin{bmatrix} 1 & 2 \\ 5 & 6 \end{bmatrix}, \quad A_{22} = \begin{bmatrix} 3 & 4 \\ 7 & 8 \end{bmatrix}$$

Similarly,

$$B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$$

Where:

$$B_{11} = \begin{bmatrix} 1 & 2 \\ 5 & 6 \end{bmatrix}, \quad B_{12} = \begin{bmatrix} 3 & 4 \\ 7 & 8 \end{bmatrix}$$
$$B_{21} = \begin{bmatrix} 9 & 10 \\ 13 & 14 \end{bmatrix}, \quad B_{22} = \begin{bmatrix} 11 & 12 \\ 15 & 16 \end{bmatrix}$$

Strassen's 7 Formulae:

Strassen's algorithm defines 7 matrix products using the submatrices:

1. $M_1 = (A_{11} + A_{22}) \times (B_{11} + B_{22})$
2. $M_2 = (A_{21} + A_{22}) \times B_{11}$
3. $M_3 = A_{11} \times (B_{12} - B_{22})$
4. $M_4 = A_{22} \times (B_{21} - B_{11})$
5. $M_5 = (A_{11} + A_{12}) \times B_{22}$
6. $M_6 = (A_{21} - A_{11}) \times (B_{11} + B_{12})$
7. $M_7 = (A_{12} - A_{22}) \times (B_{21} + B_{22})$

Each of these involves multiplying 2×2 matrices, which we'll handle **conventionally**.

Step 4: Calculate M1 to M7 (Conventional Multiplication for 2×2)

Let's compute each step-by-step.

1. Calculate $M_1 = (A_{11} + A_{22}) \times (B_{11} + B_{22})$

$$A_{11} + A_{22} = \begin{bmatrix} 5 & 10 \\ 5 & 10 \end{bmatrix} + \begin{bmatrix} 3 & 4 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 8 & 14 \\ 12 & 18 \end{bmatrix}$$

$$B_{11} + B_{22} = \begin{bmatrix} 1 & 2 \\ 5 & 6 \end{bmatrix} + \begin{bmatrix} 11 & 12 \\ 15 & 16 \end{bmatrix} = \begin{bmatrix} 12 & 14 \\ 20 & 22 \end{bmatrix}$$

Now, multiply these two 2×2 matrices conventionally:

$$M_1 = \begin{bmatrix} 8 & 14 \\ 12 & 18 \end{bmatrix} \times \begin{bmatrix} 12 & 14 \\ 20 & 22 \end{bmatrix}$$

We'll compute this multiplication next.

$$M_1 = \begin{bmatrix} 376 & 420 \\ 504 & 564 \end{bmatrix}$$

Continue this.....M2,M3,M4,M5,M6 and M7.

Step 5: Combine M_1 to M_7 to form the final result matrix

Using the Strassen combination formulas:

$$C_{11} = M_1 + M_4 - M_5 + M_7$$

$$C_{12} = M_3 + M_5$$

$$C_{21} = M_2 + M_4$$

$$C_{22} = M_1 - M_2 + M_3 + M_6$$

Final Result Matrix:

$$C = \begin{bmatrix} 450 & 500 & 550 & 600 \\ 450 & 500 & 550 & 600 \\ 90 & 100 & 110 & 120 \\ 202 & 228 & 254 & 280 \end{bmatrix}$$